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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 100102 | *Subject: Engineering Mathematics-I* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) For a Skew-Hermitian matrix A, the diagonal elements must be

(i) Real numbers
(ii) Zero or purely imaginary
(iii) Any complex number
(iv) Non-zero

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(b) If $A=\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, then A^n is

(i) $\begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$
(ii) $\begin{bmatrix} 3^n & (-4)^n \\ 1 & (-1)^n \end{bmatrix}$
(iii) $\begin{bmatrix} 1+3n & 1-4n \\ 1+n & 1-n \end{bmatrix}$
(iv) $\begin{bmatrix} 1+2n & -4n \\ 1+n & 1-2n \end{bmatrix}$

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(c) For a linear transformation $T: R^2 \times R^2$ defined by $T(x, y) = (x - y, x + y)$, then the kernel of T is

(i) R^2
(ii) $\{(0, 0)\}$ (Trivial kernal)
(iii) $\{(x, x) | x \leftarrow R\}$
(iv) $\{(x, -x) | x \leftarrow R\}$

(d) If λ be an Eigen value of the matrix A, then for the matrix $(A-\lambda I)$, which of the following statement (s) is/ are correct?

(i) Skew Symmetric
(ii) Non Singular
(iii) Singular
(iv) None of These

(e) If $f(x)=e^x$ and $g(x)=e^{-x}$ are two functions defined on $[2, 3]$ then find the value of c in the Cauchy's mean value theorem

(i) 1.5
(ii) 2.5
(iii) 2.3
(iv) 2.6

(f) The value of $\sqrt{(\frac{1}{2})}$ is

(i) $2\sqrt{\pi}$
(ii) $-2\sqrt{\frac{\pi}{2}}$
(iii) $-2\sqrt{\pi}$
(iv) $\sqrt{\frac{\pi}{2}}$

(g) If $u=\log\frac{x^2}{y}$, then $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$ is equal to

(i) $2u$
(ii) $3u$
(iii) u
(iv) 1

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(h) The integral $\int_0^a \int_x^a f(x,y)\,dy\,dx$ after changing the order of integration will be

(i) $\int_0^a \int_0^y f(x,y)\,dx\,dy$

(ii) $\int_0^a \int_y^a f(x,y)\,dx\,dy$

(iii) $\int_0^{2a} \int_0^y f(x,y)\,dx\,dy$

(iv) $\int_0^a \int_0^a f(x,y)\,dx\,dy$

(i) The value of $curl(grad\ f)$, where $f = 2x^2 - 3y^2 + 4z^2$ is is

(i) $4x - 6y + 8z$

(ii) $4x\hat{i} - 6y\hat{j} + 8z\hat{k}$

(iii) 0

(iv) 3

(j) If $\mathbf{v} = a\,x\hat{i} + b\,y\hat{j} + c\,z\hat{k}$, then the value of the integral $\int_S \mathbf{v}.d\mathbf{S}$, where S being the surface of a unit sphere, is

(i) $\frac{4\pi}{3}(a+b+c)^2$

(ii) 0

(iii) $\frac{4\pi}{3}(a+b+c)$

(iv) $\frac{4\pi}{3}(a+b+c)^3$

Q.2 (a) Find the inverse of the following matrix by the use of Gauss-Jordan method. **[6]**

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

(b) Consider following subsets of $R^3(R)$. **[4]**

(i) $W_1 = \{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$

(ii) $W_2 = \{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1\}$

Then, check whether W_1 or W_2 or both are vector subspaces of $R^3(R)$ or not.

(c) Find the dimension of the subspace of R^4 spanned by the set $\{(1, 0, 0, 0), (0, 1, 0, 0), (1, 2, 0, 1), (0, 0, 0, 1)\}$. Hence find its basis. **[4]**

Q.3 (a) For what values of λ, the system of equations $x + y + z = 1,\ x + 2y + 4z = \lambda,\ x + 4y + 10z = \lambda^2$ have solution. **[7]**

(b) Let V and W be two vector spaces in $\mathbb{R}^3$. Let $T : V \to W$ be a linear transformation defined by $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+z \\ x+y \\ x+y+z \end{bmatrix}$. Find the matrix representation of T with respect to the ordered basis **[7]**

$\mathbf{x} = \left\{\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}\right\}$ in V and $\mathbf{y} = \left\{\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}\right\}$ in W.

Q.4 (a) Check whether the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ -1 & 3 & 4 \end{bmatrix}$ is diagonalizable or not. **[6]**

(b) Write the statement of the Cayley-Hamilton theorem. If $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$, then find A^{-1} by the use of Cayley-Hamilton theorem. Also, find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$. **[8]**

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Q.5 (a) Evaluate $\underset{x\to 1}{Limit}\dfrac{x^x - x}{x - 1 - \log x}$. **[4]**

(b) Verify Rolle's Theorem for $f(x) = e^x(\sin x - \cos x)$ in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$. **[5]**

(c) Use Taylor's series to prove **[5]**

$\tan^{-1}(x+h) = \tan^{-1} x + (h\sin z)\dfrac{\sin z}{1} - (h\sin z)^2\dfrac{\sin 2z}{2} + (h\sin z)^3\dfrac{\sin 3z}{3} - \ldots$, where $z = \cot^{-1} x$.

Q.6 (a) Discuss the convergence of the improper integral $\int_a^b \dfrac{dx}{(x-a)^p}, p > 0$. **[7]**

(b) Evaluate $\int_0^1 \dfrac{x^2 dx}{\sqrt{1-x^4}} \times \int_0^1 \dfrac{dx}{\sqrt{1+x^4}}$. **[7]**

Q.7 (a) Discuss the continuity of the function $f(x,y) = \begin{cases} \dfrac{x^3 - y^3}{x^2 + y^2}, & when\ x \neq 0,\ y \neq 0 \\ 0, & when\ x = 0,\ y = 0 \end{cases}$. **[7]**

(b) Use the method of the Lagrange's multiplier to find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$. **[7]**

Q.8 (a) Change the order of integration in $\int_0^{2a}\int_{\frac{x^2}{2a}}^{\sqrt{2ax}} V\,dx\,dy$ and hence evaluate the same. **[7]**

(b) Evaluate $\iiint x^2\,y\,z\,dx\,dy\,dz$ throughout the volume bounded by the planes $x = 0,\ y = 0,\ z = 0$ and $\dfrac{x}{a} + \dfrac{y}{b} + \dfrac{z}{c} = 1$. **[7]**

Q.9 (a) Show that $div(grad\ r^n) = \nabla^2(r^n) = n(n+1)r^{n-2}$, where $r = |\mathbf{r}|$. **[6]**

(b) Evaluate $\int_S \mathbf{F}\cdot d\mathbf{S}$, where $\mathbf{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and $\mathbf{S}$ is the surface bounding the region $x^2 + y^2 = 4,\ z = 0$ and $z = 3$. **[8]**

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